COMP 10280
Programming I (Conversion)

Practical Sheet 17

Déardaoin, 15 Deireadh Fómhair 2026

For each of the following questions, write an algorithm in pseudocode first before writing a Python program. Submit your algorithms in psuedocode as well as your Python programs.
When writing functions, use one-line or multi-line docstrings, as appropriate, to document your functions.

1. Two suggested optimisations for the algorithm to calculate the divisors of a number are to initalise the divisors tuple to include 1 and the number itself and to only search from 2 and as far as half of the number. Revise the solution to include these optimisations.

   Are there further optimisations you could make?

   To clarify: the divisors of 6 are 1, 2, 3 and 6, the divisors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24, and the divisors of 7 are 1 and 7.

   Save this program as `p17p1.py`.

2. Using your solution for the previous question as a guide, implement a optimised solution for calculating the common divisors of two numbers.

   To clarify: the common divisors of 4 and 8 are 1, 2 and 4, the common divisors of 6 and 24 are 1, 2, 3 and 6, and the common divisor of 6 and 7 is 1.

   Save this program as `p17p2.py`.

3. Write a Python program that checks whether the strings "cat" and "dog" appear the same number of times in a given string input by the user.

   Save this program as `p17p3.py`.

4. Implement the recursive solution presented in lectures for checking whether given strings input by the user are palindromes.

   Save this program as `p17p4.py`.

5. Implement a variation of the palindrome program that prints out messages to allow you to trace how it works. For example, modify the `isPal` function as follows:

```
    def isPal(s):
        """Checks whether the string s is a palindrome

        Assumes that s is a str with only lowercase letters and no non-letters.
        Returns True if s is a palindrome;
        Returns False otherwise.
        Recursive function.
        Has print statements to trace its operation."""

        print('isPal function called with argument', s)
        if len(s) <= 1:
# A palindrome of length 0 or 1 is a palindrome
            print('About to return True from isPal from the base case')
            return True
        else:
# Compare the first and the last letters and check the remainder of the string
            result = s[0] == s[-1] and isPal(s[1:-1])
            print('About to return result', result, 'from isPal with argument', s)
            return result
```

Ensure you understand how this works.

Save this program as `p17p5.py`.

Please upload your work to the Brightspace site before Sunday evening.

You should keep a copy of your programs for your portfolio.